

Handles, Locks, Latches and Entry Systems



Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST3094-A	Test Light (SGT27000) or equivalent 250-300 mA incandescent bulb test lamp
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Material

Item	Specification
Motorcraft® Multi-Purpose Grease XL-5-A	ESB-M1C93-B
Motorcraft® Penetrating and Lock Lubricant (US); Motorcraft® Penetrating Fluid (Canada) XL-1 (US); CXC-51-A (Canada)	—

Principles of Operation

Power Locks

The Body Control Module (BCM) monitors the door lock control switches. When the BCM receives an unlock/lock command from either door lock control switch, the BCM energizes a relay (internal to the BCM) to supply the correct power and grounds for the door lock actuators.

Remote Keyless Entry (RKE)

The Remote Keyless Entry (RKE) feature is controlled by BCM . The BCM receives the RKE inputs from the Radio Transceiver Module (RTM) over the Medium Speed Controller Area Network (MS-CAN) . When any RKE transmitter

button is pressed, the signal is received by the RTM internal antenna. The RTM then filters the RKE signal received to make sure it is a programmed transmitter. If the signal received is from a programmed transmitter, the RTM sends the RKE command to the BCM over the MS-CAN . The BCM then interprets the RKE command and activates the corresponding output.

The BCM requests the illuminated entry feature to turn the interior lamps on when an unlock command is received. If a lock command is received, the illuminated entry feature turns off. The BCM then supplies voltage to the appropriate door lock actuator(s) to lock or unlock the doors. The RKE transmitter can also be used to activate the panic alarm. On vehicles with memory, the RKE transmitter also causes the BCM to send a MS-CAN message to the Driver Seat Module (DSM) to activate all memory features to the positions associated with the RKE transmitter being used.

Remote Start

The factory equipped remote start system is controlled by the BCM . How the BCM receives the remote start command from an Integrated Keyhead Transmitter (IKT) is very similar to the RKE function. For additional information on the factory equipped remote start system, refer to Section 303-06.

Keyless Entry Keypad

The keyless entry keypad is hardwired to the BCM . The BCM interprets the inputs from the keyless entry keypad and then controls the associated operation. The keyless entry keypad is illuminated for 5 seconds when any button is pressed. The BCM requests the illuminated entry feature to turn the interior lamps on when a valid entry code is received. If a lock all doors code is entered, the illuminated entry feature turns off.

NOTE: *The keyless entry keypad does not lock the doors if the driver door is ajar.*

The keyless entry keypad supplies a signal to the BCM when the buttons are pressed. The BCM then supplies voltage to the appropriate door lock actuator(s) to lock or unlock the doors. On vehicles with memory, the keyless entry keypad also causes the BCM to send an MS-CAN message to the DSM to activate all memory features to the positions associated with the personal entry code entered on the keypad.

Field-Effect Transistor (FET) Protection

Field-Effect Transistor (FET) is a type of transistor that when used with module software can be used to monitor and control current flow on module outputs. The FET protection strategy is used to prevent module damage in the event of excessive current flow.

The BCM utilizes an FET protective circuit strategy for many of its outputs (for example, a headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault event is detected. A short circuit DTC is stored at the fault event and a cumulative counter is started.

When the demand for the output is no longer present, the module resets the FET circuit protection to allow the circuit to function. The next time the driver requests a circuit to activate that has been shut down by a previous short (FET protection) and the circuit is still shorted, the FET protection shuts off the circuit again and the cumulative counter advances.

When the excessive circuit load occurs often enough, the module shuts down the output until a repair procedure is carried out. Each FET protected circuit has 3 predefined levels of short circuit tolerance based on the harmful effect of each circuit fault on the FET and the ability of the FET to withstand it. A module lifetime level of fault events is established based upon the durability of the FET . If the total tolerance level is determined to be 600 fault events, the 3 predefined levels would be 200, 400 and 600 fault events.

When each tolerance level is reached, the short circuit DTC that was stored on the first failure cannot be cleared by a command to clear the continuous DTCs. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful self-test proves that the fault has been repaired. After the self-test has successfully completed (no on-demand DTCs present), DTC U1000:00 and the associated DTC (the DTC related to the shorted circuit) automatically clears and the circuit function returns.

When each level is reached, the DTC associated with the short circuit sets along with DTC U1000:00. These DTCs can be cleared using the module on-demand self-test, then the Clear DTC operation on the scan tool (if the on-demand test shows

the fault corrected). The module never resets the fault event counter to zero and continues to advance the fault event counter as short circuit fault events occur.

If the number of short circuit fault events reach the third level, then DTCs U1000:00 and U3000:49 set along with the associated short circuit DTC. DTC U3000:49 cannot be cleared and the module must be replaced after the repair.

The BCM FET protected output circuit for the handles, locks, and latches system is the keypad illumination output circuit.

Inspection and Verification

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Door handles • Binding latch mechanisms	• Body Control Module (BCM) fuse(s): ▪ 18 (10A) ▪ 20 (20A) • Wiring, terminals or connectors • Door lock control switches • Keyless entry keypad • Integrated Keyhead Transmitter (IKT) • Radio Transceiver Module (RTM) • BCM

- 3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
- 4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

- 5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

- 6. If the scan tool does not communicate with the vehicle:
 - verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - Refer to Section 418-00 to diagnose no response from the PCM.

- 7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record the continuous memory DTCs for the BCM .

- 8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM .

9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B10AB:00	Remote Keyless Entry Synchronization: No Sub Type Information	GO to Pinpoint Test H.
B1218:00	Transmitter Identification Code: No Sub Type Information	GO to Pinpoint Test H.
B121A:11	Keypad Illumination Output: Circuit Short to Ground	GO to Pinpoint Test G.
B121B:01	Keypad Input Switch: General Electrical Failure	GO to Pinpoint Test F.
B126A:01	Drivers Door Interior Lock Switch: General Electrical Failure	GO to Pinpoint Test E.
B130D:00	Memory Full for Integrated Key Programming: No Sub Type Information	The maximum amount of Integrated Keyhead Transmitters (IKTs) (4) have been programmed into the Body Control Module (BCM) . Obtain all the customer's IKTs . Erase all the IKTs and program a maximum of 4.
All other DTCs	—	REFER to Section 419-10.

Symptom Chart

Symptom Chart — Electrical

Condition	Possible Causes	Action
All door locks are inoperative	- Fuse - Wiring, terminals or connectors - Body Control Module (BCM)	GO to Pinpoint Test A.
A single/more than one door lock is inoperative	- Wiring, terminals or connectors - Door lock actuators - BCM	GO to Pinpoint Test B.
The door locks operate only one way	- Wiring, terminals or connectors - Door lock control switch	GO to Pinpoint Test C.

	• BCM	
• All door locks are inoperative from one switch	• Wiring, terminals or connectors • Door lock control switch	• GO to Pinpoint Test D.
• The doors do not lock/unlock using the keyless entry keypad	• Wiring, terminals or connectors • Keyless entry keypad • BCM	• GO to Pinpoint Test F.
• The keyless entry keypad illumination is inoperative/does not operate correctly	• Wiring, terminals or connectors • Keyless entry keypad • BCM	• GO to Pinpoint Test G.
• The Remote Keyless Entry (RKE) transmitter is inoperative	• Integrated Keyhead Transmitter (IKT) • IKT battery • IKT button pressed a substantial amount of times while outside the range of the vehicle • IKT programming • Radio Transceiver Module (RTM) • BCM	• GO to Pinpoint Test H.
• An individual button/feature is inoperative from the Remote Keyless Entry (RKE) transmitter	• Door locks • Horn system • Factory remote start system (if equipped) • Turn signal system • IKT	• GO to Pinpoint Test I.
• The Remote Keyless Entry (RKE) transmitter has poor range performance	• Environmental interference • IKT • IKT battery • RTM	• GO to Pinpoint Test J.

• The autolock does not operate correctly	• Programming • Door ajar switch input • Vehicle speed signal message	• GO to Pinpoint Test K.
• The smart unlock does not operate correctly	• Key-in-ignition switch • Door ajar switch input	• GO to Pinpoint Test L.
• The auto-unlock does not operate correctly	• Programming • Door ajar switch input • Vehicle speed signal message	• GO to Pinpoint Test M.
• The memory seat does not operate correctly using the Remote Keyless Entry (RKE) transmitter	• The remote memory feature is not activated	• ACTIVATE the remote memory. REFER to Remote Memory Activation in this section. TEST the system for normal operation. If the remote memory is still inoperative, REFER to Section 501-10 to diagnose the memory seat.
• The 2-step door unlocking does not operate correctly	• The 2-step door unlocking is disabled	• PROGRAM the 2-step door unlocking. REFER to Stepped Unlock Programming in this section. TEST the system for normal operation.

Symptom Chart — Mechanical

Condition	Possible Causes	Action
• Hard to open/close door from either door handle	• Door alignment • Door hinges • Striker adjustment • Door latch	• GO to Pinpoint Test N.
• A door is difficult/does not open from the exterior door handle	• Broken or binding linkage • Exterior door handle to latch rod adjustment • Exterior door handle • Door latch	• GO to Pinpoint Test O.

• A door is difficult/does not open from the interior door handle	• Broken or binding cable/linkage • Door latch	• GO to Pinpoint Test P.
• Exterior door release handle sticks	• Binding linkage • Broken handle return spring • Exterior door handle to latch rod adjustment • Door latch	• GO to Pinpoint Test Q.
• Interior door release handle sticks	• Binding cable • Broken handle return spring • Door latch	• GO to Pinpoint Test R.
• Manual interior lock/unlock rod inoperative	• Bent or binding lock rod • Door latch	• OPERATE the door lock rod manually while observing the door lock rod for any binding or if the rod is bent. • If the rod is bent or damaged, REPAIR as necessary. TEST the system for normal operation. • If the rod is OK, INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.
• Squeak/rattle/chucking noise from door	• Door alignment • Striker adjustment • Door latch	• GO to Pinpoint Test S.
• Manual door lock cylinder inoperative	• Bent or binding lock rod and lever • Door lock cylinder • Door latch	• GO to Pinpoint Test T.
• Ignition key cannot be returned to the OFF position	• Wiring, terminals or connectors • Transmission selector lever • Binding steering column	• GO to Pinpoint Test U.

	components • Ignition lock cylinder • Ignition switch	
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Pinpoint Tests

Pinpoint Test A: All Door Locks Are Inoperative

Refer to Wiring Diagrams Cell [110](#) , Power Door Locks for schematic and connector information.

Normal Operation

The Body Control Module (BCM) supplies voltage to the lock all relay, the unlock driver door relay or unlock all relay based upon input from the door lock control switches, the Remote Keyless Entry (RKE) transmitter, or the keyless entry keypad. Upon a lock request, the BCM supplies voltage on the all door lock circuit and ground on the driver and passenger door unlock circuits. Upon an unlock request, the voltage and ground are reversed on the previously listed circuits. When the door lock and unlock relays are not energized they are connected to ground. A dedicated BCM ground circuit is provided only for the lock and unlock relays.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- BCM

PINPOINT TEST A : ALL DOOR LOCKS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK THE DOOR LOCK OPERATION FROM BOTH DOOR LOCK CONTROL SWITCHES

- Check the operation of all of the door lock actuators from both of the door lock control switches.

Are all of the door lock actuators inoperative from both door lock control switches?

Yes	GO to A2 .
No	GO to Pinpoint Test D .

A2 VERIFY ALL DOOR LOCKS ARE INOPERATIVE

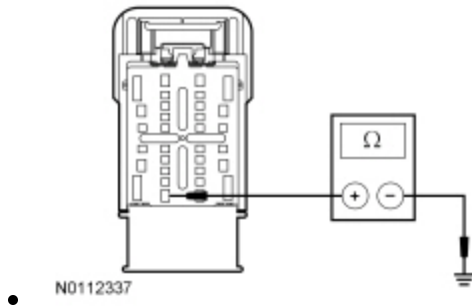
- Press the lock and unlock buttons from a door lock control switch while observing the door locks.

Are all the door locks inoperative from both switch positions (lock and unlock)?

Yes	VERIFY the BCM fuse 20 (20A) is OK. If OK, GO to A3 . If not OK, GO to A5 .
No	CHECK the symptom(s) and GO to Symptom Chart — Electrical .

A3 CHECK THE POWER DOOR LOCK RELAY GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: BCM [C2280D](#) .
- Measure the resistance between the BCM [C2280D](#) Pin 16, circuit GD139 (BK/YE), harness side and ground.

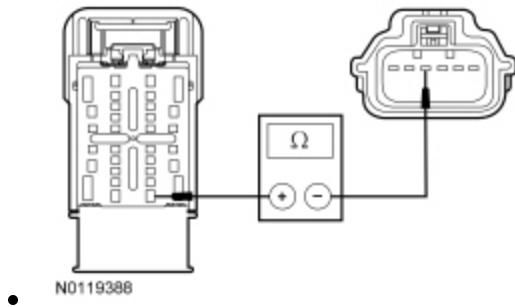


Is the resistance less than 5 ohms?

Yes	GO to A4 .
No	REPAIR circuit GD139 (BK/YE) for an open. TEST the system for normal operation.

A4 CHECK THE ALL LOCK CIRCUIT FOR AN OPEN

- Disconnect: LH Front Door Lock Actuator [C525](#).
- Measure the resistance between the BCM [C2280D](#) Pin 26, circuit CPL11 (GY/BN), harness side and the LH front door lock actuator [C525](#) Pin 4, circuit CPL11 (GY/BN), harness side.



Is the resistance less than 5 ohms?

Yes	GO to A6 .
No	REPAIR CPL11 (GY/BN) for an open. TEST the system for normal operation.

A5 CHECK THE ALL LOCK CIRCUIT FOR A SHORT TO GROUND

- Install a new BCM fuse 20 (20A).
- Operate the door lock actuators by pressing LOCK on the door lock control switch.

Is the BCM fuse 20 (20A) OK?

Yes	REPAIR circuit CPL51 (BU/GN) or circuit CPL52 (VT/GY) for a short to ground. TEST the system for normal operation.
No	REPAIR circuit CPL11 (GY/BN) for a short to ground. TEST the system for normal operation.

A6 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: A Single/More Than One Door Lock Is Inoperative

Refer to Wiring Diagrams Cell [110](#) , Power Door Locks for schematic and connector information.

Normal Operation

The Body Control Module (BCM) supplies voltage to the lock all relay, the unlock driver door relay or unlock all relay based upon input from the door lock control switches, the Remote Keyless Entry (RKE) transmitter, or the keyless entry keypad. Upon a lock request, the **BCM** supplies voltage on the all door lock circuit and ground on the driver and passenger door unlock circuits. Upon an unlock request, the voltage and ground are reversed on the previously listed circuits. When the door lock and unlock relays are not energized they are connected to ground. A dedicated **BCM** ground circuit is provided only for the lock and unlock relays.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Door lock actuators
- **BCM**

PINPOINT TEST B : A SINGLE/MORE THAN ONE DOOR LOCK IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTICE: The following Pinpoint Test uses a test light to simulate normal circuit loads. Use only the test lamp recommended in the Special Tools table at the beginning of this section or equivalent. To avoid connector terminal damage, use the Rotunda Flex Probe kit for the test light probe connection to the vehicle. Do not use the test lamp probe directly on any connector.

B1 CHECK THE DOOR LATCH FOR BINDING

- Lock and unlock the inoperative door lock using the door lock rod.

Does the door lock and unlock?

Yes	GO to B3 .
No	GO to B2 .

B2 CHECK THE LOCK ROD FOR BINDING

- Operate the door lock rod manually while observing the door lock rod for any binding or if the rod is bent.

Is the door lock rod bent or binding?

Yes	REPAIR as necessary. TEST the system for normal operation.
No	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.

B3 CHECK THE ALL LOCK CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Disconnect: Inoperative Door Lock Actuator .
- Connect a test light between the inoperative door lock actuator, harness side and ground as follows:

Inoperative Door Lock Actuator	Connector-Pin	Circuit
LH front	C525 Pin 4	CPL11 (GY/BN)
RH front	C609 Pin 4	CPL11 (GY/BN)
LH rear	C705 Pin 4	CPL11 (GY/BN)
RH rear	C805 Pin 4	CPL11 (GY/BN)



• N0120890

- **NOTE:** The BCM only supplies voltage to the actuator momentarily. It is important to monitor the test light while pressing the door lock control switch.

Monitor the test light while pressing the door lock control switch in the LOCK position.

Does the test light momentarily illuminate?

Yes	GO to B4 .
No	REPAIR circuit CPL11 (GY/BN) for an open. TEST the system for normal operation.

B4 CHECK THE UNLOCK CIRCUITS FOR VOLTAGE

- Connect a test light between the inoperative door lock actuator, harness side and ground as follows:

Inoperative Door Lock Actuator	Connector-Pin	Circuit
LH front	C525 Pin 3	CPL51 (BU/GN)
RH front	C609 Pin 3	CPL52 (VT/GY)
LH rear	C705 Pin 3	CPL52 (VT/GY)
RH rear	C805 Pin 3	CPL52 (VT/GY)



• N0120891

- **NOTE:** The BCM only supplies voltage to the actuator momentarily. It is important to monitor the test light while pressing the door lock control switch.

Monitor the test light while pressing the door lock control switch in the UNLOCK position.

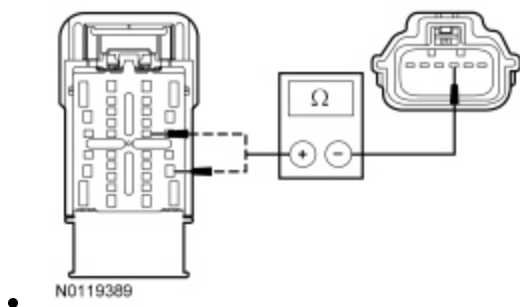
Does the test light momentarily illuminate?

Yes	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.
No	GO to B5 .

B5 CHECK THE UNLOCK CIRCUITS FOR AN OPEN

- Disconnect: BCM [C2280D](#) .
- Measure the resistance between the BCM , harness side and the inoperative door lock actuator, harness side as follows:

BCM Connector-Pin	Inoperative Door Lock Actuator Connector-Pin	Circuit
C2280D Pin 21	C525 Pin 3	CPL51 (BU/GN)
C2280D Pin 31	C609 Pin 3	CPL52 (VT/GY)
C2280D Pin 31	C705 Pin 3	CPL52 (VT/GY)
C2280D Pin 31	C805 Pin 3	CPL52 (VT/GY)



Is the resistance less than 5 ohms?

Yes	GO to B6 .
No	REPAIR circuit CPL51 (BU/GN) or circuit CPL52 (VT/GY) for an open. TEST the system for normal operation.

B6 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: The Door Locks Operate Only One Way

Refer to Wiring Diagrams Cell [110](#) , Power Door Locks for schematic and connector information.

Normal Operation

The Body Control Module (BCM) sends voltage signals to the door lock control switches through the lock and unlock input circuits. When the lock or unlock switch is pressed, the corresponding voltage signal is routed to ground, signaling the **BCM** to lock or unlock the doors. The door lock control switches have a dedicated ground circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Door lock control switch

- BCM

PINPOINT TEST C : THE DOOR LOCKS OPERATE ONLY ONE WAY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

C1 CHECK THE RECORDED BCM DTCS FROM THE SELF-TEST

- Check the recorded results from the BCM self-test.

Is DTC B126A:01 present?

Yes	GO to Pinpoint Test E.
No	GO to C2 .

C2 VERIFY THE OPERATION OF BOTH DOOR LOCK CONTROL SWITCHES

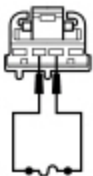
- Ignition OFF.
- Press the lock and unlock button of both door lock control switches while observing the door lock operation.

Do the door locks operate only one way from both switches?

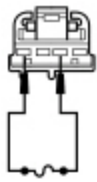
Yes	GO to C4 .
No	GO to C3 .

C3 CHECK THE LOCK INPUT OR UNLOCK INPUT CIRCUIT FOR AN OPEN (SINGLE SWITCH INOPERATIVE)

- Disconnect: Suspect Door Lock Control Switch .
- Unlock the doors from the working door lock control switch.
- Connect a fused jumper wire between the LH door lock control switch [C505](#) Pin 3, circuit CPK19 (BU/GN), harness side and the LH door lock control switch [C505](#) Pin 2, circuit GD134 (BK/VT), harness side; or between the RH door lock control switch [C605](#) Pin 3, circuit CPK19 (BU/GN), harness side and the RH door lock control switch [C605](#) Pin 2, circuit GD139 (BK/YE), harness side.



- N0072779
- Remove the jumper wire.
- Connect a fused jumper wire between the LH door lock control switch [C505](#) Pin 4, circuit CPK23 (YE/VT), harness side and the LH door lock control switch [C505](#) Pin 2, circuit GD134 (BK/VT), harness side; or between the RH door lock control switch [C605](#) Pin 4, circuit CPK23 (YE/VT), harness side and the RH door lock control switch [C605](#) Pin 2, circuit GD139 (BK/YE), harness side.



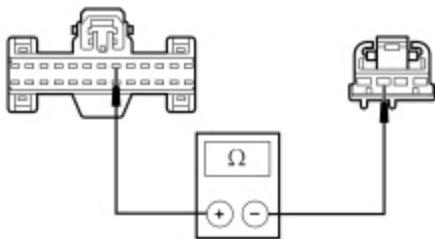
• N0090257

Do the doors lock and unlock with the jumper wire?

Yes	REMOVE the jumper wire. INSTALL a new door lock control switch. REFER to Door Lock Control Switch in this section. TEST the system for normal operation.
No	REMOVE the jumper wire. REPAIR the input circuit in question. TEST the system for normal operation.

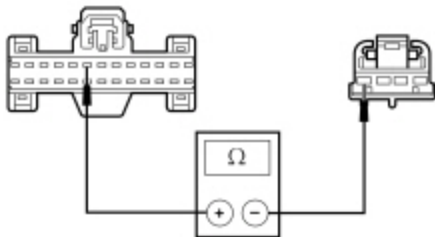
C4 CHECK THE LOCK INPUT OR UNLOCK INPUT CIRCUIT FOR AN OPEN (BOTH SWITCHES INOPERATIVE)

- Disconnect: RH Door Lock Control Switch [C605](#).
- Disconnect: BCM [C2280C](#) .
- If the doors do not lock, measure the resistance between the BCM [C2280C](#) Pin 6, circuit CPK19 (BU/GN), harness side and the RH door lock control switch [C605](#) Pin 3, circuit CPK19 (BU/GN), harness side.



• N0120802

- If the doors do not unlock, measure the resistance between the BCM [C2280C](#) Pin 8, circuit CPK23 (YE/VT), harness side and the RH door lock control switch [C605](#) Pin 4, circuit CPK23 (YE/VT), harness side.



• N0120803

Is the resistance less than 5 ohms?

Yes	GO to C5 .
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No	REPAIR circuit CPK19 (BU/GN) or circuit CPK23 (YE/VT) for an open. TEST the system for normal operation.
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C5 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: All Door Locks Are Inoperative From One Switch

Refer to Wiring Diagrams Cell [110](#) , Power Door Locks for schematic and connector information.

Normal Operation

The Body Control Module (BCM) sends voltage signals to the door lock control switches through the lock and unlock input circuits. When the lock or unlock switch is pressed, the corresponding voltage signal is routed to ground, signaling the **BCM** to lock or unlock the doors. The door lock control switches have a dedicated ground circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Door lock control switch

PINPOINT TEST D : ALL DOOR LOCKS ARE INOPERATIVE FROM ONE SWITCH

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

D1 CHECK THE SUSPECT DOOR LOCK CONTROL SWITCH FOR CORRECT OPERATION

- Disconnect: Suspect Door Lock Control Switch .
- Carry out the door lock control switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Does the door lock control switch pass the component test?

Yes	REPAIR circuit GD134 (BK/VT) or GD139 (BK/YE) for an open. TEST the system for normal operation.
No	INSTALL a new door lock control switch. REFER to Door Lock Control Switch in this section. TEST the system for normal operation.

Pinpoint Test E: DTC B126A:01

Refer to Wiring Diagrams Cell [110](#) , Power Door Locks for schematic and connector information.

Normal Operation

The Body Control Module (BCM) sends voltage signals to the door lock control switches through the lock and unlock input circuits. When the lock or unlock switch is pressed, the corresponding voltage signal is routed to ground, signaling the BCM to lock or unlock the doors. The door lock control switches have a dedicated ground circuit.

- DTC B126A:01 (Drivers Door Interior Lock Switch: General Electrical Failure) — sets when the BCM detects a short to ground from the lock or unlock request input circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Door lock control switch
- BCM

PINPOINT TEST E : DTC B126A:01

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
E1 CHECK THE RECORDED BCM DTCS FROM THE SELF-TEST	
• Check the recorded results from the BCM self-test.	
Is DTC B126A:01 present?	
Yes	GO to E2 .
No	CHECK the symptom(s) and GO to Symptom Chart — Electrical .
E2 CHECK THE LH DOOR LOCK CONTROL SWITCH	
• Disconnect: LH Door Lock Control Switch C505. • Repeat the BCM self-test.	
Is DTC B126A:01 retrieved again?	
Yes	LEAVE the switch disconnected. GO to E3 .
No	INSTALL a new LH door lock control switch. REFER to Door Lock Control Switch in this section. TEST the system for normal operation.

E3 CHECK THE RH DOOR LOCK CONTROL SWITCH

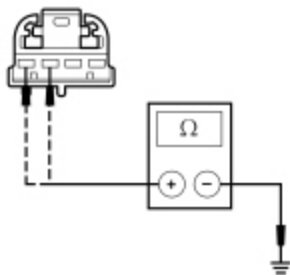
- Disconnect: RH Door Lock Control Switch [C605](#).
- Repeat the BCM self-test.

Is DTC B126A:01 retrieved again?

Yes	LEAVE the switch disconnected. GO to E4 .
No	INSTALL a new RH door lock control switch. REFER to Door Lock Control Switch in this section. TEST the system for normal operation.

E4 CHECK THE LOCK OR UNLOCK INPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the RH door lock control switch [C605](#) Pin 3, circuit CPK19 (BU/GN), harness side and ground; and between the RH door lock control switch [C605](#) Pin 4, circuit CPK23 (YE/VT), harness side and ground.



N0120804

Are the resistances greater than 10,000 ohms?

Yes	GO to E5 .
No	REPAIR circuit CPK19 (BU/GN) or circuit CPK23 (YE/VT) for a short to ground. CLEAR the DTCs. REPEAT the self-test.

E5 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test F: The Doors Do Not Lock/Unlock Using The Keyless Entry Keypad

Refer to Wiring Diagrams Cell [117](#) , Remote Keyless Entry and Alarm for schematic and connector information.

Normal Operation

The Body Control Module (BCM) monitors for a ground signal on the keyless entry keypad signal input circuits. When an individual keypad button is pressed, an individual or combination of the input circuits is routed to ground. The BCM then determines which button was pressed based on which circuit(s) are grounded. The keypad has a dedicated ground circuit. Once the BCM determines that the 5-digit personal entry code, or the factory set 5-digit entry code has been entered on the keypad, the BCM then energizes the driver door unlock relay to unlock the driver door (stepped unlock enabled).

To unlock all the doors, the 3/4 button must be pressed within 5 seconds of the 5-digit code being entered.

To lock all the doors, the 5-digit code does not need to be entered. Press the 7/8 and 9/0 buttons at the same time, with the driver door closed.

The keyless entry keypad has an anti-scan feature to help prevent theft. If a valid 5-digit code is not entered within 7 attempts (35 consecutive button presses), the keyless entry keypad goes into anti-scan. In anti-scan mode, the keypad illumination flashes and all input is ignored except for pressing the 7/8 and 9/0 buttons, which will lock all doors.

- DTC B121B:01 (Keypad Input Switch: General Electrical Failure) — sets when the BCM detects a short to ground on any of the keyless entry keypad input circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Keyless entry keypad
- BCM

PINPOINT TEST F : THE DOORS DO NOT LOCK/UNLOCK USING THE KEYLESS ENTRY KEYPAD

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
F1 CHECK THE BCM KEYPAD CODE	
• Enter the following diagnostic mode on the scan tool: BCM DataLogger . • Observe the BCM PIDs (KEYCODE_1stD, KEYCODE_2ndD, KEYCODE_3rdD, KEYCODE_4thD, and KEYCODE_5thD). Read and record the 5-digit factory set entry code. • Enter the 5-digit factory set entry code on the keyless entry keypad.	
Do the doors unlock?	
Yes	The system is operating as designed. INFORM the customer of the correct system operation.
No	GO to F2 .

F2 RETRIEVE THE RECORDED BCM DTCS FROM THE SELF-TEST

- Check for recorded BCM DTCS from the self-test.

Is DTC B121B:01 recorded?

Yes	GO to F3 .
No	GO to F4 .

F3 REPEAT THE BCM SELF-TEST

- Disconnect: Keyless Entry Keypad [C500](#).
- Clear the DTCS, then repeat the BCM self-test.

Is DTC B121B:01 recorded?

Yes	GO to F9 .
No	INSTALL a new keyless entry keypad. REFER to Keyless Entry Keypad in this section. TEST the system for normal operation.

F4 CHECK THE BCM KEYPAD PIDS

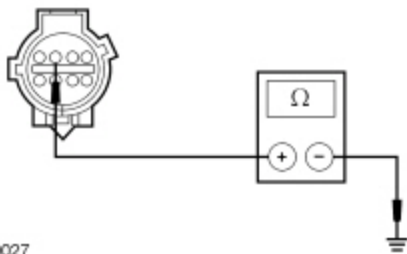
- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Press each keyless entry keypad button while observing the PIDs (KYPD_BTN_1/2, KYPD_BTN_3/4, KYPD_BTN_5/6, KYPD_BTN_7/8, and KYPD_BTN_9/0).

Do the PIDs display active when the corresponding button is pressed?

Yes	GO to F10 .
No	GO to F5 .

F5 CHECK THE KEYPAD GROUND CIRCUIT FOR AN OPEN

- Disconnect: Keyless Entry Keypad [C500](#).
- Measure the resistance between the keyless entry keypad [C500](#) Pin 3, circuit GD134 (BK/VT), harness side and ground.



• A0070027

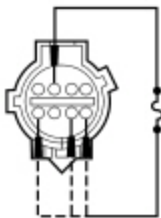
Is the resistance less than 5 ohms?

Yes	GO to F6 .
No	REPAIR circuit GD134 (BK/VT) for an open. TEST the system for normal operation.

F6 CHECK THE KEYLESS ENTRY KEYPAD

- Connect a fused jumper wire between the keyless entry keypad, harness side, while monitoring the BCM PIDs (KYPD_BTN_1/2, KYPD_BTN_7/8, and KYPD_BTN_9/0) as follows:

Keyless Entry Keypad Connector- Pin/Circuit	Keyless Entry Keypad Connector- Pin/Circuit	BCM PID
C500 Pin 8 CPK29 (GY/BU)	C500 Pin 3 GD134 (BK/VT)	KYPD_BTN_1/2
C500 Pin 6 CPK30 (VT/GN)	C500 Pin 3 GD134 (BK/VT)	KYPD_BTN_7/8
C500 Pin 5 CPK31 (YE/GN)	C500 Pin 3 GD134 (BK/VT)	KYPD_BTN_9/0



• N0104320

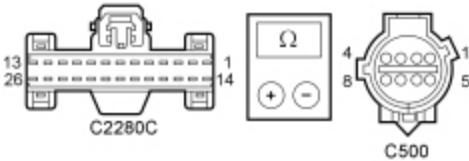
Do the PIDs display active when the jumper wire is connected?

Yes	INSTALL a new keyless entry keypad. REFER to Keyless Entry Keypad in this section.
No	REMOVE the jumper wire. GO to F7 .

F7 CHECK THE KEYPAD INPUT CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the BCM , harness side and the keyless entry keypad, harness side as follows:

BCM Connector-Pin	Circuit	Keyless Entry Keypad Connector-Pin
C2280C Pin 24	CPK29 (GY/BU)	C500 Pin 8
C2280C Pin 18	CPK30 (VT/GN)	C500 Pin 6
C2280C Pin 11	CPK31 (YE/GN)	C500 Pin 5



- N0120805

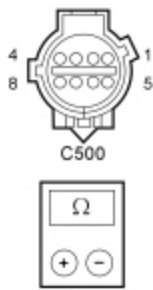
Are the resistances less than 5 ohms?

Yes	GO to F8 .
No	REPAIR the circuit in question for an open. TEST the system for normal operation.

F8 CHECK THE KEYPAD INPUT CIRCUITS FOR A SHORT TO EACH OTHER

- Measure the resistance between the keyless entry keypad, harness side as follows:

Keyless Entry Keypad Connector-Pin	Circuits	Keyless Entry Keypad Connector-Pin
C500 Pin 8	CPK29 (GY/BU) and CPK30 (VT/GN)	C500 Pin 6
C500 Pin 8	CPK29 (GY/BU) and CPK31 (YE/GN)	C500 Pin 5
C500 Pin 6	CPK30 (VT/GN) and CPK31 (YE/GN)	C500 Pin 5



N0120806

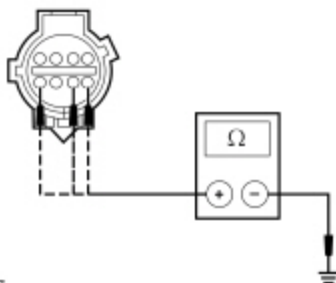
Are the resistances greater than 10,000 ohms?

Yes	GO to F9 .
No	REPAIR the circuits in question. TEST the system for normal operation.

F9 CHECK THE KEYPAD INPUT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the keyless entry keypad, harness side and ground as follows:

Connector-Pin	Circuit
C500 Pin 8	CPK29 (GY/BU)
C500 Pin 6	CPK30 (VT/GN)
C500 Pin 5	CPK31 (YE/GN)



N0120807

Are the resistances greater than 10,000 ohms?

Yes	GO to F10 .
No	REPAIR the circuit in question for a short to ground. CLEAR the DTCs. REPEAT the self-test.

F10 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:

- corrosion
- damaged pins
- pushed-out pins

- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test G: The Keyless Entry Keypad Illumination Is Inoperative/Does Not Operate Correctly

Refer to Wiring Diagrams Cell [117](#) , Remote Keyless Entry and Alarm for schematic and connector information.

Normal Operation

When any keypad button is pressed the Body Control Module (BCM) sends voltage to the keyless entry keypad LEDs on the illumination circuit to assist the driver in entering an access code at night. The keyless entry keypad has a separate ground circuit for the LEDs.

The keyless entry keypad has an anti-scan feature to help prevent theft. If a valid 5-digit code is not entered within 7 attempts (35 consecutive button presses), the keyless entry keypad goes into anti-scan mode. In anti-scan mode, the keypad illumination flashes and all input is ignored except for pressing the 7/8 and 9/0 buttons, which locks all the doors.

- DTC B121A:11 (Keypad Illumination Output: Circuit Short to Ground) — sets when the BCM detects a short to ground on the keyless entry keypad illumination circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Keyless entry keypad
- BCM

PINPOINT TEST G : THE KEYLESS ENTRY KEYPAD ILLUMINATION IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

G1 CHECK THE KEYLESS ENTRY KEYPAD OPERATION

- Press the 7/8 and 9/0 buttons simultaneously on the keyless entry keypad.

Do all of the doors lock?

Yes	GO to G2 .
No	GO to Pinpoint Test F .

G2 CHECK THE RECORDED BCM DTCS FROM THE SELF-TEST

- Check the recorded results from the BCM self-test.

Is DTC B121A:11 present?

Yes	GO to G9 .
No	GO to G3 .

G3 CHECK THE KEYLESS ENTRY KEYPAD ILLUMINATION

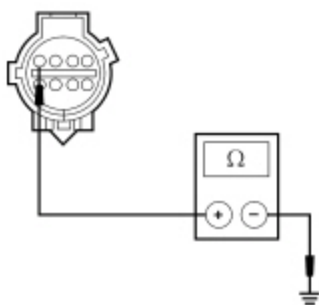
- Check the keyless entry keypad illumination.

Is the keyless entry keypad illumination always on?

Yes	GO to G8 .
No	GO to G4 .

G4 CHECK THE KEYPAD ILLUMINATION GROUND CIRCUIT FOR AN OPEN

- Disconnect: Keyless Entry Keypad [C500](#).
- Measure the resistance between the keyless entry keypad [C500](#) Pin 4, circuit GD134 (BK/VT), harness side and ground.



• N0090260

Is the resistance less than 5 ohms?

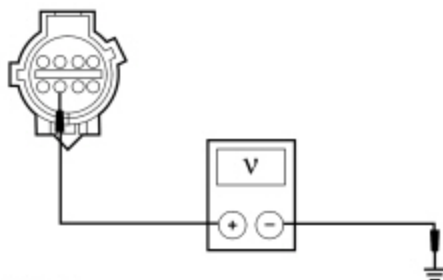
Yes	GO to G5 .
No	REPAIR circuit GD134 (BK/VT) for an open. CLEAR the DTCs. REPEAT the self-test.

G5 CHECK THE KEYPAD ILLUMINATION CIRCUIT FOR REFERENCE VOLTAGE

NOTE: The BCM sends out a constant reference voltage on circuit CPK28 (WH/GN) for its own circuit diagnostics. This voltage is 10 volts (+/- 1 volt).

- Ignition ON.

- Measure the voltage between the keyless entry keypad [C500](#) Pin 7, circuit CPK28 (WH/GN), harness side and ground.



• A0078888

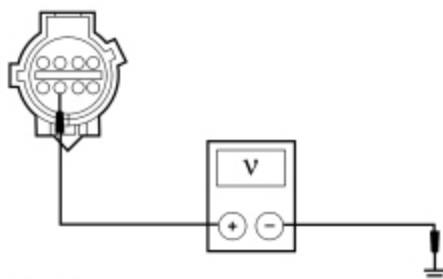
Is 10 volts (+/- 1 volt) present?

Yes	GO to G6 .
No	GO to G7 .

G6 CHECK FOR BCM OUTPUT

NOTE: This pinpoint test checks for a voltage increase above the reference voltage checked in the previous pinpoint test when the keypad lamp is active commanded on.

- Enter the following diagnostic mode on the scan tool: **BCM DataLogger** .
- Measure the voltage between the keyless entry keypad [C500](#) Pin 7, circuit CPK28 (WH/GN), harness side and ground while active commanding the **BCM** PID (KEY_PAD_LMP) on.



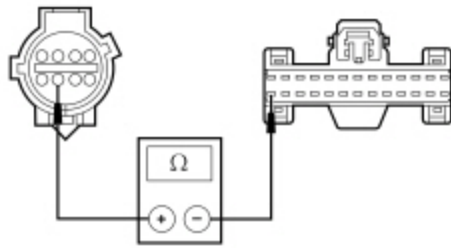
• A0078888

Is the voltage greater than 11 volts?

Yes	INSTALL a new keyless entry keypad. REFER to Keyless Entry Keypad in this section. CLEAR the DTCs. REPEAT the self-test.
No	GO to G11 .

G7 CHECK THE KEYPAD ILLUMINATION CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: **BCM** [C2280C](#) .
- Measure the resistance between the keyless entry keypad [C500](#) Pin 7, circuit CPK28 (WH/GN), harness side and the **BCM** [C2280C](#) Pin 26, circuit CPK28 (WH/GN), harness side.



N0120808

Is the resistance less than 5 ohms?

Yes	GO to G11 .
No	REPAIR circuit CPK28 (WH/GN) for an open. CLEAR the DTCs. REPEAT the self-test.

G8 CHECK THE KEYPAD ILLUMINATION CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: BCM [C2280C](#) .
- Ignition ON.
- Observe the keypad illumination.

Is the keypad still illuminated?

Yes	REPAIR circuit CPK28 (WH/GN) for a short to voltage. CLEAR the DTCs. REPEAT the self-test.
No	GO to G11 .

G9 CHECK THE KEYLESS ENTRY KEYPAD FOR A SHORT TO GROUND

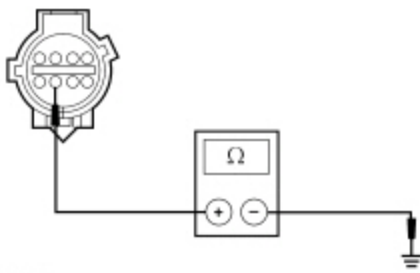
- Disconnect: Keyless Entry Keypad [C500](#).
- Clear the DTCs, then repeat the BCM self-test.

Is DTC B121A:11 recorded?

Yes	GO to G10 .
No	INSTALL a new keyless entry keypad. REFER to Keyless Entry Keypad in this section. TEST the system for normal operation.

G10 CHECK THE KEYPAD ILLUMINATION CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the keyless entry keypad [C500](#) Pin 7, circuit CPK28 (WH/GN), harness side and ground.



N0058740

Is the resistance greater than 10,000 ohms?

Yes	GO to G11 .
No	REPAIR circuit CPK28 (WH/GN) for a short to ground. CLEAR the DTCs. REPEAT the self-test.

G11 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: The Remote Keyless Entry (RKE) Transmitter Is Inoperative

Normal Operation

The Remote Keyless Entry (RKE) feature is controlled by the Radio Transceiver Module (RTM) and the Body Control Module (BCM) . When a button is pressed on an Integrated Keyhead Transmitter (IKT) the Transmitter Identification Code (TIC) and **RKE** command is received by the internal RTM antenna. If the RTM determines that it is a valid and programmed **TIC** , it sends the **RKE** command and **TIC** to the **BCM** over the Medium Speed Controller Area Network (MS-CAN) . When the **BCM** receives the **RKE** command, it carries out the command by controlling the door locks or activating the horn and turn signals for the panic alarm.

The **RKE** transmitters and the **BCM** also utilize a rolling code to prevent the code from being "captured" by a code grabber. The system advances the counter in the **RKE** transmitter and the **BCM** every time an **RKE** transmitter button is pressed.

- DTC B10AB: 00 (Remote Keyless Entry Synchronization: No Sub Type Information) — sets when the **BCM** detects the rolling counter received from an **RKE** transmitter is out of synchronization with the rolling counter stored in the

module.

- DTC B1218: 00 (Transmitter Identification Code: No Sub Type Information) — sets when the BCM detects an invalid TIC from an IKT .

This pinpoint test is intended to diagnose the following:

- IKT
- IKT battery
- IKT button pressed a substantial amount of times while outside the range of the vehicle
- IKT programming
- RTM
- BCM

PINPOINT TEST H : THE RKE TRANSMITTER IS INOPERATIVE

NOTE: Aftermarket or dealer-installed systems may adversely affect the RKE system operation. These systems should be disconnected before diagnosing any RKE concerns.

H1 CHECK FOR THE CORRECT IKTS

- **NOTE:** Make sure the IKTs are those provided with the OEM system and not from an aftermarket system, or a dealer-installed system that may have been installed on the vehicle.

Check that the correct IKTs are used with the vehicle. Compare the IKTs with the one shown in Handles, Locks, Latches and Entry Systems in the Description and Operation portion of this section.

Are the correct RKE transmitters present?

Yes	GO to H2 .
No	The system cannot be tested without the correct IKTs . INFORM the customer that the correct IKTs must be present to proceed with diagnosis of the system.

H2 CHECK THE RKE OPERATION WHEN THE RTM IS IN AWAKE MODE

- Open and close the driver door.
- Press any button on the IKT .

Does the RKE feature operate?

Yes	GO to H3 .
No	GO to H4 .

H3 CHECK THE RKE OPERATION WHEN THE RTM IS IN SLEEP MODE

- Ignition OFF.
- Let the vehicle sit for 10 minutes to allow the vehicle modules to go into a sleep mode.
- Press any button on the IKT .

Does the RKE feature operate?

Yes	The system is operating correctly at this time.
-----	---

No	GO to H12 .
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H4 PERFORM THE NETWORK TEST ON THE RTM

- Connect the scan tool.
- Ignition ON.
- Perform the network test on the RTM.

Does the RTM pass the network test?

Yes	GO to H5 .
No	REFER to Section 418-00 to diagnose the RTM does not respond to the scan tool.

H5 CHECK FOR TPMS DTCS

- Perform the BCM self-test.

Are any TPMS DTCS recorded?

Yes	DIAGNOSE all TPMS DTCS first. REFER to section Section 204-04.
No	GO to H6 .

H6 CHECK FOR DTC B10AB:00 OR DTC B1218:00 FROM THE BCM SELF-TEST

- Check the recorded results from the BCM self-test.

Is DTC B10AB:00 or DTC B1218:00 present?

Yes	For DTC B10AB: 00 GO to H7 . For DTC B1218: 00 GO to H8 .
No	GO to H8 .

H7 RESYNCHRONIZE THE INOPERATIVE IKT

- Ignition OFF.
- Ignition ON.
- Ignition OFF.
- Press any button on the IKT .

Does the RKE feature operate?

Yes	The system is OK. CLEAR the DTCS. REPEAT the self-test.

No	GO to H8 .
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H8 MAKE SURE THE RKE TRANSMITTER SIGNAL IS BEING RECEIVED

- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Press a button on the **IKT** while observing the **BCM** PIDs (TIC_LAST_BC and LST_RKE_xMTR).

Does the Transmitter Identification Code (TIC) and RKE command show up on the scan tool screen when a button is pressed?

Yes	GO to H9 .
No	GO to H10 .

H9 CHECK IF THE RKE TRANSMITTERS ARE PROGRAMMED

- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Press a button on the **IKT** and compare the **BCM** PID (TIC_LAST_BC) with the PIDs (TIC_01_BCM, 2, 3, 4, 7, 8, 9, or 10).

Does the TIC displayed match any of the TICs stored in memory?

Yes	GO to H13 .
No	PROGRAM the inoperative IKT individually or PROGRAM all the IKTs . REFER to Section 419-01B for the programming procedures. If all the IKTs are to be programmed, INFORM the customer that any IKTs not present need to be programmed. CLEAR the DTCs. REPEAT the self-test.

H10 CHECK THE IKT BATTERY

- Using a thin coin, open the **IKT** .
- Do not clean off any grease from the battery terminals on the back surface of the circuit board.
- Verify the correct battery is used (CR2032).
- Remove the **IKT** battery and measure the voltage.

Is the voltage greater than 2.5 volts?

Yes	GO to H11 .
No	INSTALL a new battery (make sure the battery is seated correctly). DO NOT reprogram the IKT (weak or dead batteries do not erase TICs from memory). TEST the system for normal operation.

H11 CHECK FOR NORMAL OPERATION WITH A KNOWN GOOD IKT

- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .

- Use the customer's second **IKT** , or a known good **RKE** transmitter that is correct for the vehicle, press a button on the **IKT** while observing the **BCM** PID (TIC_LAST_BC).

Does the **TIC show up on the scan tool when a button is pressed?**

Yes	REPLACE the inoperative IKT and PROGRAM the new IKT individually or PROGRAM all the IKT s . REFER to Section 419-01B for the programming procedures. If all the IKT s are to be programmed, INFORM the customer that any IKT s not present need to be programmed. CLEAR the DTCs. REPEAT the self-test.
No	GO to H13 .

H12 CHECK FOR CORRECT RTM OPERATION

- Disconnect the RTM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the RTM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new RTM (TPM module). REFER to Section 204-04. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

H13 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test I: An Individual Button/Feature Is Inoperative From The Remote Keyless Entry (RKE) Transmitter

Normal Operation

The Remote Keyless Entry (RKE) feature is controlled by the Radio Transceiver Module (RTM) and the Body Control Module (BCM) . When a button is pressed on an Integrated Keyhead Transmitter (IKT) the Transmitter Identification Code (TIC) and RKE command is received by the internal RTM antenna. If the RTM determines that it is a valid and programmed TIC , it sends the RKE command and TIC to the BCM over the Medium Speed Controller Area Network (MS-CAN) . When the BCM receives the RKE command, it carries out the command by controlling the door locks or activating the horn and turn signals for the panic alarm.

The BCM will disable the panic feature if the ignition key is in any position other than OFF or LOCK.

This pinpoint test is intended to diagnose the following:

- Door locks
- Factory remote start system (if equipped)
- Horn system
- Turn signal system
- IKT

PINPOINT TEST I : AN INDIVIDUAL BUTTON/FEATURE IS INOPERATIVE FROM THE RKE TRANSMITTER

I1 VERIFY THE DOOR LOCK OPERATION

- Press the lock and unlock button from the driver door lock control switch while observing the door lock operation.

Do the door locks operate correctly?

Yes	GO to I2 .
No	CHECK the symptom(s) and GO to Symptom Chart — Electrical .

I2 VERIFY THE HORN OPERATION

- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Select the BCM PID (HORN) and active command the horn on.

Does the horn sound when commanded on?

Yes	GO to I3 .
No	REFER to Section 413-06.

I3 VERIFY THE HAZARD LAMP OPERATION

- Place the hazard switch in the ON position.

Do the hazard lamps operate correctly?

Yes	For vehicles without remote start, REPLACE the inoperative IKT and PROGRAM the new IKT individually or PROGRAM all the IKT s . REFER to Section 419-01B for the programming procedures. If all the IKT s are to be programmed, INFORM the customer that any IKT s not present need to be programmed. TEST the system for normal operation. For vehicles equipped with the factory remote start, GO to 14 .
No	REFER to Section 417-01.

I4 VERIFY THE REMOTE START BUTTON

- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Select the **BCM** PID (LST_RKE_xMTR) while pressing the remote start button on an **IKT** .

Does the PID indicate the remote start button is being pressed?

Yes	REFER to Section 303-06.
No	REPLACE the inoperative IKT and PROGRAM the new IKT individually or PROGRAM all the IKT s . REFER to Section 419-01B for the programming procedures. If all the IKT s are to be programmed, INFORM the customer that any IKT s not present need to be programmed. TEST the system for normal operation.

Pinpoint Test J: The Remote Keyless Entry (RKE) Transmitter Has Poor Range Performance

Normal Operation

The Remote Keyless Entry (RKE) transmitter has a normal operating range of 10 m (33 ft). The Radio Transceiver Module (RTM) has an internal antenna to receive the **RKE** transmitter signal. When the RTM receives a command from a programmed **RKE** transmitter, the RTM sends the **RKE** signal to the Body Control Module (BCM) , which processes the command to provide the appropriate output.

This pinpoint test is intended to diagnose the following:

- Environmental interference
- **IKT**
- **IKT** battery
- RTM

PINPOINT TEST J : THE RKE TRANSMITTER HAS POOR RANGE PERFORMANCE

NOTE: All **RKE** transmitters must be present to begin diagnosis of the **RKE** system.

NOTE: Aftermarket or dealer-installed systems can adversely affect the **RKE** system operation. These systems must be disconnected before diagnosing any **RKE** concerns.

J1 CHECK FOR THE CORRECT RKE TRANSMITTERS

- **NOTE:** Make sure the **RKE** transmitters are those provided with the OEM system and not from an aftermarket system, or a dealer-installed system that may have been installed on the vehicle.

Check that the correct **IKT**s are used with the vehicle. Compare the **IKT**s with the one shown in Handles, Locks, Latches and Entry Systems in the Description and Operation portion of this section.

Are the correct **RKE transmitters present?**

Yes	GO to J2 .
No	The system cannot be tested without the correct RKE transmitters. INFORM the customer that the correct RKE transmitters must be present to proceed with diagnosis of the system.

J2 CHECK ALL RKE TRANSMITTERS FOR POOR RANGE PERFORMANCE

- **NOTE:** *The 10 m (33 ft) measurement of range is not the standard but is a guideline that clearly indicates a vehicle is experiencing normal range performance.*

Check all **RKE** transmitters for poor range performance (less than 10 m [33 ft]).

Do all **RKE transmitters experience poor range?**

Yes	GO to J4 .
No	GO to J3 .

J3 CHECK THE IKT BATTERY

- Using a thin coin, open the **IKT** .
- Do not clean off any grease from the battery terminals on the back surface of the circuit board.
- Verify the correct battery is used (CR2032).
- Remove the **IKT** battery and measure the voltage.

Is the voltage greater than 2.5 volts?

Yes	REPLACE the inoperative IKT and PROGRAM the new IKT individually or PROGRAM all the IKTs . REFER to Section 419-01B for the programming procedures. If all the IKTs are to be programmed, INFORM the customer that any IKTs not present need to be programmed. TEST the system for normal operation.
No	INSTALL a new battery (make sure the battery is seated correctly). DO NOT reprogram the IKT (weak or dead batteries do not erase TICs from memory). TEST the system for normal operation.

J4 CHECK THE LOCATION OF THE VEHICLE AND THE APPROACH ANGLES AROUND THE VEHICLE

- Make sure the poor performance is consistent in nature and is not from one approaching angle.
- The **RKE** transmitter range performance may be degraded in certain locations. For example, if the vehicle is within 0.5 mile (0.8 km) of high-power devices or radio/TV towers, the operating distance of the **RKE** transmitters may be reduced.

Is the poor range performance consistent around the vehicle?

Yes	GO to J5 .
No	The system is operating correctly at this time. The range may have been affected by environmental

interference.

J5 CHECK FOR CORRECT RTM OPERATION

- Disconnect all the RTM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the RTM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new RTM (TPM module). REFER to Section 204-04. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: The Autolock Does Not Operate Correctly

Normal Operation

The Body Control Module (BCM) energizes the lock relay based on input from the ajar switches, the ignition switch, the Transmission Range (TR) sensor and the vehicle speed signal. The BCM locks the doors when the following conditions are met:

- All the doors are closed
- The ignition switch is in the RUN position
- The vehicle is shifted into any gear putting the vehicle in motion
- The vehicle attains a speed greater than 20 km/h (12 mph)

This pinpoint test is intended to diagnose the following:

- Programming
- Door ajar switch input
- Vehicle speed signal message

PINPOINT TEST K : THE AUTOLOCK DOES NOT OPERATE CORRECTLY

K1 CHECK THE COURTESY LAMP OPERATION

- Open and close the doors while checking the courtesy lamp operation.

Do the courtesy lamps operate correctly?

Yes	GO to K2 .
No	REFER to Section 417-02.

K2 CHECK THE VEHICLE SPEED OPERATION

- Check the operation of the speedometer.

Does the speedometer operate correctly?

Yes	ENABLE the autolock feature. REFER to Autolock and Auto-Unlock Programming in this section.
No	REFER to Section 413-01.

Pinpoint Test L: The Smart Unlock Does Not Operate Correctly

Normal Operation

The Body Control Module (BCM) commands all doors to unlock based on input from the key-in-ignition switch and the door ajar signal. This feature helps prevent the keys from being locked in the vehicle by unlocking the doors if the key is in the ignition and either front door is ajar.

The vehicle can still be locked with the key in the ignition, by using the manual lock rod, a key in the door lock cylinder, the keyless entry keypad, or the LOCK button on a Integrated Keyhead Transmitter (IKT) .

This pinpoint test is intended to diagnose the following:

- Key-in-ignition switch
- Door ajar switch input

PINPOINT TEST L : THE SMART UNLOCK DOES NOT OPERATE CORRECTLY

L1 CHECK THE DOOR LOCK OPERATION

- Check the operation of the door locks from the driver door lock control switch.

Do all the doors lock and unlock?

Yes	GO to L2 .
No	CHECK the symptom(s) and GO to Symptom Chart — Electrical .

L2 CHECK THE COURTESY LAMP OPERATION

- Open and close the left front door while checking the courtesy lamp operation.

Do the courtesy lamps operate correctly?

Yes	REFER to Section 413-01 to DIAGNOSE the key-in-ignition warning chime.
No	REFER to Section 417-02.

Pinpoint Test M: The Auto-Unlock Does Not Operate Correctly

Normal Operation

The Body Control Module (BCM) energizes the unlock relay based on input from the door ajar switches and the vehicle speed signal. The BCM unlocks the doors when the following conditions are met:

- The vehicle speed has exceeded 20 km/h (12 mph) then dropped to 0 km/h (0 mph)
- The key is turned to the OFF or ACC position
- The LH front door is opened within 10 minutes of the key being turned to the OFF or ACC position

The auto-unlock feature is cancelled if the doors are locked with the door lock control switch or the Integrated Keyhead Transmitter (IKT) within the 10-minute active window. After the 10-minute active window expires, the auto-unlock feature does not unlock the doors if the driver door is opened.

This pinpoint test is intended to diagnose the following:

- Programming
- Door ajar switch input
- Vehicle speed signal message

PINPOINT TEST M : THE AUTO-UNLOCK DOES NOT OPERATE CORRECTLY

M1 CHECK THE COURTESY LAMP OPERATION	
• Open and close the doors while observing the courtesy lamp operation.	
Do the courtesy lamps operate correctly?	
Yes	GO to M2 .
No	REFER to Section 417-02.

M2 CHECK THE VEHICLE SPEED OPERATION	
• Check the operation of the speedometer.	
Does the speedometer operate correctly?	
Yes	ENABLE the auto-unlock feature. REFER to Autolock and Auto-Unlock Programming in this section.
No	REFER to Section 413-01.

Pinpoint Test N: Hard To Open/Close Door From Either Door Handle

Normal Operation

The door latch can be actuated from the interior or exterior door handle. When actuated, the door latch releases and allows the door to open. If the door latch or the door hinges have insufficient lubrication or if the striker or door are misaligned, it causes extra force to be used when opening or closing the door.

This pinpoint test is intended to diagnose the following:

- Door alignment

- Door hinges
- Striker adjustment
- Door latch

PINPOINT TEST N : HARD TO OPEN/CLOSE DOOR FROM EITHER DOOR HANDLE

N1 CHECK THE LATCH OPERATION FROM BOTH DOOR HANDLES

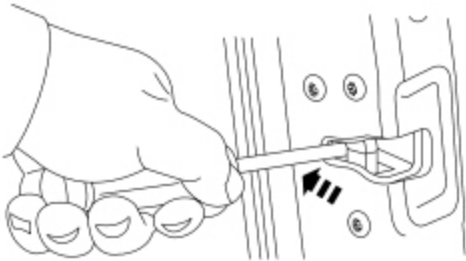
- Open and close the door using both the interior and exterior door handles.

Does the door open normally from one of the door handles?

Yes	If the door is difficult/does not open from the exterior door handle, GO to Pinpoint Test O . If the door is difficult/does not open from the interior door handle, GO to Pinpoint Test P .
No	If the door does not operate correctly from both door handles, GO to N2 .

N2 CHECK THE LATCH OPERATION

- Open the door.
- Using a screwdriver, fully close the latch (2 clicks).



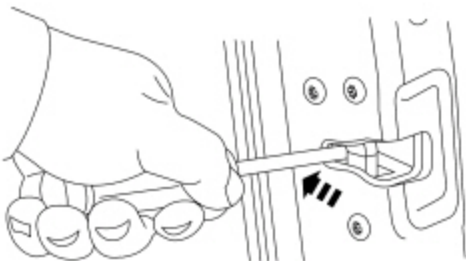
- N0094195
- Verify that the latch releases easily while pulling on the interior/exterior door handle.

Does the latch release easily?

Yes	GO to N4 .
No	GO to N3 .

N3 CHECK THE LATCH OPERATION AFTER LUBRICATION

- Lubricate the door latch. Refer to Latch Lubrication in this section.
- Using a screwdriver, fully close the latch (2 clicks).



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- Verify that the latch releases easily while pulling on the interior/exterior door handle.

Does the latch release easily?

Yes	The concern was caused by an insufficiently lubricated door latch.
No	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.

N4 CHECK THE STRIKER ADJUSTMENT

- Check the adjustment of the striker. Refer to Section 501-03.

Is the striker adjusted correctly?

Yes	GO to N5 .
No	ADJUST the striker as necessary. TEST the system for normal operation.

N5 CHECK THE DOOR ALIGNMENT

- Check the alignment of the door. Refer to Section 501-03.

Is the door aligned correctly?

Yes	LUBRICATE the door hinges. TEST the system for normal operation.
No	ADJUST the door as necessary. TEST the system for normal operation.

Pinpoint Test O: A Door Is Difficult/Does Not Open From The Exterior Door Handle

Normal Operation

The exterior door handle is connected to the door latch with an actuating rod. When the exterior door handle is pulled it causes the actuating rod to push down on the latch lever. The actuating rod can be adjusted for proper exterior door handle operation. When the latch lever is moved, the door latch releases, allowing the door to open.

This pinpoint test is intended to diagnose the following:

- Broken or binding linkage
- Exterior door handle to latch rod adjustment
- Exterior door handle
- Door latch

PINPOINT TEST O : A DOOR IS DIFFICULT/DOES NOT OPEN FROM THE EXTERIOR DOOR HANDLE

O1 CHECK THE LATCH OPERATION FROM BOTH DOOR HANDLES

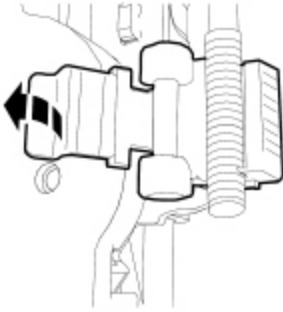
- Open the door using both the interior and exterior door handles.

Does the door open normally from one of the door handles?

Yes	If the door is difficult/does not open from only the exterior door handle, GO to O2 .
No	If the door does not operate correctly from both door handles, GO to Pinpoint Test N .

O2 CHECK THE EXTERIOR DOOR HANDLE

- Remove the door trim panel. Refer to Section 501-05.
- Open the clip and disconnect the exterior door handle actuating rod.



- Pull and release the exterior door handle.

Does the exterior door handle operate correctly?

Yes	GO to O3 .
No	INSTALL a new exterior door handle. REFER to Exterior Front Door Handle or Exterior Door Handle — SuperCrew in this section. TEST the system for normal operation.

O3 CHECK THE EXTERIOR DOOR HANDLE AND LINKAGE OPERATION

- Operate the exterior door handle while observing the linkage.

Are any of the exterior door handle components or linkages binding or broken?

Yes	REPAIR as necessary. TEST the system for normal operation.
No	GO to O4 .

O4 CARRY OUT THE EXTERIOR DOOR HANDLE ADJUSTMENT

- Carry out the exterior door handle to latch rod adjustment. Refer to Door Handle to Latch Rod Adjustment in this section.

Does the exterior door handle operate correctly?

Yes	The concern was caused by an improperly adjusted exterior door handle.
No	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.

Pinpoint Test P: A Door Is Difficult/Does Not Open From The Interior Door Handle

Normal Operation

The interior door handle is connected to the door latch by an actuating cable. When the interior door handle is pulled it causes the cable to pull on the latch lever. When the latch lever is moved, the door latch releases, allowing the door to open.

This pinpoint test is intended to diagnose the following:

- Broken or binding cable/linkage
- Door latch

PINPOINT TEST P : A DOOR IS DIFFICULT/DOES NOT OPEN FROM THE INTERIOR DOOR HANDLE

P1 CHECK THE LATCH OPERATION FROM BOTH DOOR HANDLES

- For the rear SuperCrew doors, make sure the child lock is in the unlock position.
- Open the door using both the interior and exterior door handles.

Does the door open normally from one of the door handles?

Yes	If the door is difficult/does not open from only the interior door handle, GO to P2 .
No	If the door does not operate correctly from both door handles, GO to Pinpoint Test N .

P2 CHECK THE INTERIOR DOOR HANDLE AND CABLE/LINKAGE OPERATION

- Remove the door trim panel. Refer to Section 501-05.
- Install the interior door handle back onto the release cable.
- Operate the interior door handle while observing the cable/linkage.

Are any of the interior door handle components or cable/linkages binding or broken?

Yes	REPAIR as necessary. TEST the system for normal operation.
No	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.

Pinpoint Test Q: Exterior Door Release Handle Sticks

Normal Operation

The exterior door handle is connected to the door latch with an actuating rod. When the exterior door handle is pulled it causes the actuating rod to push down on the latch lever. The actuating rod can be adjusted for proper exterior door handle operation. When the latch lever is moved, the door latch releases, allowing the door to open. The handle has a return spring to make sure the handle returns to a closed position.

This pinpoint test is intended to diagnose the following:

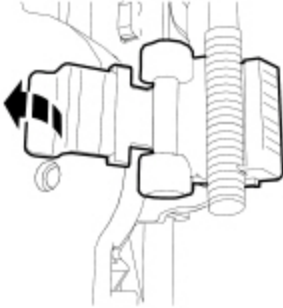
- Binding linkage

- Broken handle return spring
- Exterior door handle to latch rod adjustment
- Door latch

PINPOINT TEST Q : EXTERIOR DOOR RELEASE HANDLE STICKS

Q1 CHECK FOR A BROKEN RETURN SPRING

- Remove the door trim panel. Refer to Section 501-05.
- Open the clip and disconnect the exterior door handle actuating rod.



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- Pull and release the exterior door handle.

Does the exterior door handle return to a closed position once released?

Yes	GO to Q2 .
No	INSTALL a new exterior door handle. REFER to Exterior Front Door Handle or Exterior Door Handle — SuperCrew in this section. TEST the system for normal operation.

Q2 CARRY OUT THE EXTERIOR DOOR HANDLE ADJUSTMENT

- Carry out the exterior door handle to latch rod adjustment. Refer to Door Handle to Latch Rod Adjustment in this section.

Does the exterior door handle operate correctly?

Yes	The concern was caused by an improperly adjusted exterior door handle.
No	GO to Q3 .

Q3 CHECK THE EXTERIOR DOOR HANDLE AND LINKAGE OPERATION

- Operate the exterior door handle while observing the linkage.

Are any of the exterior door handle components or linkages binding?

Yes	REPAIR as necessary. TEST the system for normal operation.
No	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.

Pinpoint Test R: Interior Door Release Handle Sticks

Normal Operation

The interior door handle is connected to the door latch by an actuating cable. When the interior door handle is pulled it causes the cable to pull on the latch lever. When the latch lever is moved, the door latch releases, allowing the door to open. The handle has a return spring to make sure the handle returns to a closed position.

This pinpoint test is intended to diagnose the following:

- Binding cable
- Broken handle return spring
- Door latch

PINPOINT TEST R : INTERIOR DOOR RELEASE HANDLE STICKS

R1 CHECK FOR A BROKEN RETURN SPRING	
• Remove the interior door handle. Refer to Interior Door Handle — Front or Interior Door Handle — SuperCrew in this section. • Pull and release the interior door handle.	
Does the interior door handle return to a closed position once released?	
Yes	GO to R2 .
No	INSTALL a new interior door handle. REFER to Interior Door Handle — Front or Interior Door Handle — SuperCrew in this section. TEST the system for normal operation.

R2 CHECK THE INTERIOR DOOR HANDLE CABLE OPERATION	
• Remove the door latch release cable. • Observe the door latch release cable for any damage or binding.	
Is the door latch release cable OK?	
Yes	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.
No	REPAIR as necessary. TEST the system for normal operation.

Pinpoint Test S: Squeak/Rattle/Chucking Noise From Door

This pinpoint test is intended to diagnose the following:

- Door alignment
- Striker adjustment
- Door latch

PINPOINT TEST S : SQUEAK/RATTLE/CHUCKING NOISE FROM DOOR

S1 CHECK FOR ANY LOOSE COMPONENTS	
• Remove the door trim panel. Refer to Section 501-05.	

- Inspect inside the door for any loose components.

Are there any loose components inside the door?

Yes	REPAIR as necessary. TEST the system for normal operation.
No	GO to S2 .

S2 CHECK THE STRIKER ADJUSTMENT

- Check the adjustment of the striker. Refer to Section 501-03.

Is the striker adjusted correctly?

Yes	GO to S3 .
No	ADJUST the striker as necessary. TEST the system for normal operation.

S3 CHECK THE DOOR ALIGNMENT

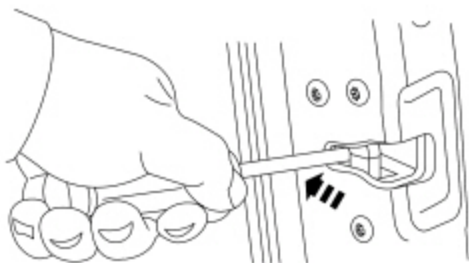
- Check the alignment of the door. Refer to Section 501-03.

Is the door aligned correctly?

Yes	GO to S4 .
No	ADJUST the door as necessary. TEST the system for normal operation.

S4 CHECK THE LATCH OPERATION AFTER LUBRICATION

- Lubricate the door latch. Refer to Latch Lubrication in this section.
- Using a screwdriver, fully close the latch (2 clicks).



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- Operate the door latch and listen for the noise.

Is the original noise still present after the latch is lubricated?

Yes	INSTALL a new door latch. REFER to Front Door Latch or Rear Door Latch — SuperCrew in this section. TEST the system for normal operation.
No	The concern was caused by an insufficiently lubricated door latch.

Pinpoint Test T: Manual Door Lock Cylinder Inoperative

Normal Operation

The door lock cylinder is connected to the door latch via a lock rod and can be used to manually lock/unlock a door.

This pinpoint test is intended to diagnose the following:

- Bent or binding lock rod and lever
- Door lock cylinder
- Door latch

PINPOINT TEST T : MANUAL DOOR LOCK CYLINDER INOPERATIVE

T1 CHECK THE LOCK CYLINDER OPERATION AFTER LUBRICATION

- Spray a lock lubricant such as Motorcraft XL-1 into the lock cylinder opening for a couple seconds.
- Operate the door lock cylinder with the key.

Does the door lock and unlock using the door lock cylinder?

Yes	The concern was caused by an insufficiently lubricated lock cylinder. Spray a multi-purpose grease such as Motorcraft XL-5 into the lock cylinder for a couple seconds to provide long term lubrication.
No	GO to T2 .

T2 CHECK THE DOOR LOCK CYLINDER

- Remove the door lock cylinder. Refer to Door Lock Cylinder in this section.
- Operate the door lock cylinder with the key.

Does the door lock cylinder rotate freely to both lock and unlock positions?

Yes	GO to T3 .
No	INSTALL a new door lock cylinder. REFER to Door Lock Cylinder in this section. TEST the system for normal operation.

T3 CHECK THE DOOR LOCK CYLINDER LINKAGE

- Install the door lock cylinder. Refer to Door Lock Cylinder in this section.
- Operate the door lock cylinder while observing the linkage.

Is the door lock cylinder rod and lever bent or binding?

Yes	REPAIR as necessary. TEST the system for normal operation.

No	INSTALL a new door latch. REFER to Front Door Latch in this section. TEST the system for normal operation.
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Pinpoint Test U: Unable To Remove The Key From The Ignition Lock Cylinder

Refer to Wiring Diagrams Cell [37](#) , Shift Interlock for schematic and connector information.

Normal Operation

The ignition switch contains a key removal inhibit solenoid. The solenoid prevents the ignition lock cylinder to be turned to lock and the key removed any time the transmission selector lever is in any position except park. The inhibit solenoid receives voltage at all times from the Body Control Module (BCM) fuse 28 (15A). When the transmission selector lever is in any gear except park, the selector switch closes, which supplies ground to the solenoid.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Transmission selector lever
- Binding steering column components
- Ignition lock cylinder
- Ignition switch

PINPOINT TEST U : IGNITION KEY CANNOT BE RETURNED TO THE OFF POSITION

U1 CHECK FOR AN ENERGIZED KEY REMOVAL INHIBIT SOLENOID

- Remove the BCM fuse 28 (15A).
- Attempt to turn the ignition key to the run position and then back to the lock position and remove the key.

Does the ignition lock cylinder turn to the lock position and can the key be removed?

Yes	REFER to Section 211-05 to continue diagnosis of the key removal inhibit solenoid system.
No	GO to U2 .

U2 CHECK THE IGNITION LOCK CYLINDER FOR BINDING

- Remove the ignition lock cylinder. Refer to Ignition Lock Cylinder — Functional in this section.
- Rotate the ignition lock cylinder through all positions using the key.

Does the ignition lock cylinder freely rotate through all positions and can the key be removed from the lock cylinder?

Yes	REFER to Section 211-05 to continue diagnosis of the steering column for any binding steering column components or ignition switch.
No	INSTALL a new ignition lock cylinder. REFER to Ignition Lock Cylinder — Functional in this section. TEST the system for normal operation.